# Functional Specification Document (FSD) – Logistics ERP Analytics

## 1. Overview

This document describes the functional requirements of the Logistics ERP Analytics system. It defines how the system will operate, user interactions, and expected outputs.

## 2. System Scope

The system will process logistics, inventory, supplier, and shipment data to generate dashboards, reports, and predictive insights using Power BI.

## 3. User Roles

- CEO: View high-level KPIs
- CFO: Financial insights
- COO: Operations monitoring
- Warehouse Manager: Inventory tracking
- Procurement Manager: Supplier analysis
- Logistics Manager: Shipment tracking

## 4. Functional Modules

- Order Management
- Inventory Management
- Procurement & Supplier Analysis
- Logistics & Shipment Tracking
- Reporting & Dashboarding

## 5. Detailed Functional Requirements

Order Management:
- Track Order ID, Date, Quantity, Value
- Monitor order fulfillment

Inventory Management:
- Track opening, received, issued, closing stock
- Identify stock-outs and overstock

Procurement:

- Analyze supplier performance
- Track GRN and lead times

Logistics:
- Track shipments, couriers, delivery status
- Calculate delivery time

Reporting:
- Generate dashboards and KPIs
- Enable filtering and drill-down

## 6. Input Data

Customer, Product, Supplier, Order, Shipment, Inventory fields from dataset.

## 7. Output Reports

- CEO Dashboard
- CFO Dashboard
- COO Dashboard
- Warehouse Dashboard
- Procurement Dashboard
- Logistics Dashboard

## 8. KPIs

- Delivery Time
- On-Time Delivery Rate
- Inventory Turnover
- Supplier Performance
- Stock-out Rate

## 9. User Interaction

Users will interact via Power BI dashboards with filters, slicers, drill-down, and drill-through features.

## 10. Constraints

- Data accuracy depends on source system
- Performance depends on data volume
- Power BI limitations on large datasets